

Developing a Utility-Scale Solar Farm in the UK: From Greenfield to EPC Signature

Executive summary

A typical UK utility-scale solar farm — say a 49.9 MW project under the Town and Country Planning Act, or a 100 MW+ Nationally Significant Infrastructure Project (NSIP) under the Development Consent Order regime — takes roughly **4–7 years from greenfield site identification to EPC contract signature** and absorbs **£1.5–4.5 million of at-risk development capital ($\approx$ £30,000–£90,000 per MW)** before financial close. Land control, planning consent, an accepted grid connection offer and an offtake / route-to-market arrangement are the four non-negotiable gates that must be in place before an EPC contract can be signed and project finance drawn.

The journey is now being reshaped by three structural reforms that any developer must internalise:

1. **NSIP threshold raised from 50 MW to 100 MW** (Infrastructure Planning (Onshore Wind and Solar Generation) Order 2025), in force from **31 December 2025** with a one-year transitional window — moving the planning “dead zone” up from 49.9 MW to ~99.9 MW.
2. **TMO4+ Connections Reform** (Ofgem-approved 15 April 2025; “live” from 10 June 2025): the historic “first come, first served” queue is replaced with a “first ready, first connected” gated process administered by **NESO** (the National Energy System Operator, which took over from National Grid ESO in October 2024). Around 482 GW of the existing 596 GW queue is expected to drop to Gate 1 or be terminated, with the new Gate 2 queue sized to roughly 296 GW aligned to Clean Power 2030/2035 capacities.
3. **Clean Power 2030 Action Plan** (December 2024) targets **45–57 GW of solar by 2030** (from ~18.1 GW installed in March 2025), with **mandatory 10% Biodiversity Net Gain (BNG)** applying since February 2024 [Solar Power Portal](#) and AR7a in early 2026 awarding a record **4.9 GW of solar at a strike price of £65/MWh (2024 prices)**. [PV Tech](#)

Industry analysts now most-cited in UK solar development decisions include Solar Energy UK, RenewableUK, Cornwall Insight (PPA tracking and route-to-market revenue forecasting), LCP Delta (CfD bid strategy and wholesale market modelling), Pexapark and LevelTen (European PPA price indices), and Aurora Energy Research (Clean Power 2030 stress-testing). DESNZ commissioned Arup's July 2025 Renewable Energy Generation Cost

and Technical Assumptions 2024 study [Publishing Service](#) as the official cost baseline for AR7 and AR8.

The report below walks through each milestone of the development journey, with current UK-specific figures, gatekeeping requirements and common failure points.

1. Site identification and land control

How developers find sites

The three primary screening criteria are:

- **Grid proximity:** sites within roughly 1–10 miles (and ideally <1 mile) of a 33 kV, 66 kV, 132 kV or 400 kV connection point. Conrad Energy's George Hall describes "less than 10 miles from an electricity substation" [Farmers Weekly](#) as the principal critical test. Above ~50 MW, a transmission (132 kV/400 kV) connection via NESO is needed; below ~50 MW, a distribution (DNO) connection at 33 kV or 132 kV is typical.
- **Solar irradiance:** irradiance in southern England (Kent, Wiltshire, Hampshire, Cornwall, Suffolk) is materially higher than in northern England and Scotland, and lease rents reflect this — Savills and Farmers Weekly both note "lower rents in the north of the country where solar radiation levels are lower." [Myerson](#)
- **Land type and constraints:** open, gently sloping (preferably south-facing) land, away from designated landscapes (National Parks, AONBs/National Landscapes), heritage assets, ancient woodland, SSSIs, flood zones 2/3, the Green Belt (now requiring "very special circumstances" for sub-100 MW projects under the December 2024 NPPF revision), and major aviation safeguarding zones.

Agricultural Land Classification (ALC) and Best and Most Versatile (BMV) land

Land is graded 1 (excellent) to 5 (very poor). Grades 1, 2 and 3a constitute **Best and Most Versatile (BMV) agricultural land**. [Solargridcheck](#) National Policy Statement EN-3 (updated in 2024) does not prohibit solar on BMV land, but developers must demonstrate that poorer-quality land cannot accommodate the project. **Projects affecting more than 20 hectares of BMV land must consult Natural England**. In practice, most developers and landowners target **Grade 3b, 4 and 5** land (RWE, for instance, explicitly states "preferably we are looking for grade 3 land or worse"). [Rwe](#) The December 2024 NPPF revision removed a previous provision allowing LPAs to refuse solar on food-security grounds, but local sensitivity remains acute and a robust site-specific ALC survey is now near-universal.

Mis-classification of land at planning stage — or relying on the 1980s/1990s national ALC maps rather than a site-specific ALC survey — is one of the most common reasons for refusal.

Site size: MW-per-acre rules of thumb

Industry rule-of-thumb in the UK is **2–2.5 acres per MWp** ($\approx 0.8\text{--}1\text{ ha/MWp}$), with modern bifacial single-axis tracker designs at the lower end. ILOS Energy’s published case-study scaling is “1 MW per 2.4 acres” — so a 50 MW project typically takes ~ 120 acres ($\sim 48\text{ ha}$), a 100 MW project ~ 250 acres ($\sim 100\text{ ha}$) and the largest UK NSIPs (e.g. the 500 MW Sunnica, 600 MW Cottam, 400 MW Longfield, 500 MW Heckington Fen) extend to 1,000+ ha. As of September 2024, UK ground-mount solar covered $\sim 21,200$ hectares — about 0.1% of UK land area.

Option agreements, leases and Heads of Terms

A developer almost always secures land through a two-stage structure:

Instrument	Typical duration	Typical financial terms (2025/26)
Exclusivity / Heads of Terms	1–3 months	Often nil or nominal (£1k–£10k)
Option Agreement	2–5 years (typically 3 years + 1–2 year extensions)	Option fee £100–£500 per acre per year , Solargridcheck sometimes paid as a lump sum at signing; developer covers all planning/grid/legal costs
Lease (triggered on planning consent)	25–40 years , Solargridcheck Solargridcheck sometimes 50 years; renewal options common	£850–£1,350 per acre per year Solargridcheck Solargridcheck (Savills, Farmers Weekly, SolarGridCheck 2026), indexed to RPI/CPI. Alternative: 5–6% of turnover Farmers Weekly (“top-up” mechanism), worth $\sim$ £1,400–£1,500/acre at current power prices on a good site Farmers Weekly

A **decommissioning bond of £15,000–£25,000 per MW** [Solargridcheck](#) is now standard market practice, sometimes held by the developer’s funder or via an on-balance-sheet provision. Section 106 agreements with the LPA frequently include a landscape and ecology management plan obligation. [Winacc](#)

Heads of Terms are usually 8–15 pages and cover: option period and extension fees; rent

(acreage vs turnover hybrid); permitted use; insolvency / step-in for lenders; sub-letting and assignment to a SPV at financial close; reinstatement and decommissioning bond; tenancy security excluded from Landlord & Tenant Act 1954 Part II; and tax indemnities (particularly Agricultural Property Relief and Business Property Relief).

Key documents and deliverables at this stage

Site shortlist memo; constraints map (GIS overlay of grid, irradiance, ALC, designations, flood, heritage, ecology); Heads of Terms; Option Agreement; agreed form of Lease; landowner consultation log.

Common failure points at this stage

- Landlord pulling out before option exercise (no security of tenure).
 - “Clustering” disputes where multiple developers chase the same grid node and overlapping landholdings.
 - Failure to control land for cable corridor, substation footprint and BNG off-site units — increasingly a problem as off-site BNG can require additional 15–30% land area.
[Solargridcheck](#)
 - The High Court has recently confirmed (2024) that developing a project adjacent to an existing solar farm may be deemed an “extension” — pushing combined capacity over the NSIP threshold [Dentons](#) and forcing the project into the DCO regime (“Beware the slip into NSIP”).
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2. Grid connection

Institutional framework

- **NESO (National Energy System Operator)** — operational since October 2024, replacing National Grid ESO. NESO is publicly owned and reports to DESNZ/Ofgem. It runs the transmission queue and the new Gate 1/Gate 2 process.
- **Transmission Owners (TOs)** — National Grid Electricity Transmission (NGET) in England & Wales, SP Transmission (Scottish Power) in southern Scotland, SSEN Transmission in northern Scotland.
- **Distribution Network Operators (DNOs)** — UK Power Networks, Northern Powergrid, Electricity North West, SP Distribution, SSEN Distribution, National Grid Electricity Distribution (formerly WPD).

- **Ofgem** regulates network charges and approves connection code modifications (CMPs).

TNUoS, BSUoS and DUoS — the charging framework

Generators pay **Transmission Network Use of System (TNUoS)** charges set by NESO and administered by NGET, varying by zone — historically **strongly negative (a credit) in southern England** and strongly positive (a cost) in northern Scotland, encouraging generation to locate near demand. Reforms under TCR (Targeted Charging Review) and now the Strategic Spatial Energy Plan (SSEP) work are flattening these signals. **Balancing Services Use of System (BSUoS)** has, since April 2023, been paid by suppliers only (final demand), eliminating one major historical cost for generators. Distribution-connected projects also pay (or receive) **Distribution Use of System (DUoS)** charges, with embedded benefits eroded by Targeted Charging Review changes phased in from 2018.

TMO4+ Connections Reform — the most important change for solar developers

Ofgem approved the **TMO4+ ("First Ready, First Connected") reform package on 15 April 2025**, and the new regime went live on **10 June 2025**. Key features:

- **Two gates:**
 - **Gate 1:** indicative offer only — no confirmed connection date, no firm queue position, no connection point. For projects that do not meet readiness or strategic-alignment criteria.
 - **Gate 2:** firm offer with confirmed connection date, point and queue position. Requires that the project be **"ready"** (secured land rights, sufficiently advanced planning) and **"needed"** (aligned with Clean Power 2030 Action Plan capacity envelopes for the relevant technology, zone and decade).
- **Gate 2 to Whole Queue (G2TWQ) process:** all existing offer-holders (~596 GW) had to re-evidence by **26 August 2025** [National Energy System Operator](#) (extended from 29 July). NESO estimated ~482 GW will be pushed to Gate 1 [Womble Bond Dickinson](#) or terminated, leaving a firm Gate 2 queue of ~296 GW (including built capacity) aligned with CPAP capacities to 2035. [Womble Bond Dickinson](#)
- **Revised offer timeline** (NESO update, December 2025): Gate 2 Phase 1 transmission/large-embedded offers between **mid-May 2026 and mid-September 2026**; Gate 2 Phase 1 distribution offers between **early July 2026 and mid-November 2026**; Phase 2 transmission offers between **early September 2026 and mid-January 2027**; Phase 2 distribution between **mid-October 2026 and mid-March 2027**. This is a 4–5 month delay versus NESO's original timetable.

- **Annual application windows** thereafter (at minimum) — the next window after the August 2025 closure is not expected before mid-2026, frustrating developers who landed in Gate 1.
- **Connections Action Plan (CAP)** — the joint Ofgem/DESNZ document published in late 2023, which sets the policy framework for TMO4+ and the ENA's three-point plan on queue management.

Readiness criteria for Gate 2 (solar)

Broadly: (i) demonstrable land rights — option, lease or freehold over the project site; (ii) submitted planning application **or** planning permission already granted, depending on project size; (iii) committed funding for connection-stage securities; (iv) red-line boundary defined; (v) for transmission-connected projects, a Transmission Impact Assessment commenced.

Connection costs, fees and securities

Item	Typical figure (2025/26)
Initial connection application fee (DNO / NESO)	£5,000–£50,000 depending on size
Grid studies (System Impact Study, Connection Impact Assessment, ER P28/G99 protection studies)	£30,000–£200,000
TNUoS (Transmission Network Use of System) — generators	Zone-dependent (can be a credit in southern England, a charge in northern Scotland). Highly material for project IRR.
Connection construction cost	Typically £150,000–£500,000 per MW for a 33 kV DNO connection; up to £1–3 million per MW for transmission-connected projects requiring deep reinforcement. The Connections Charging Review (2023) shifted

(developer-funded works)	more “shallow-ish” reinforcement costs back onto users.
Securities	Liability cap securities (bank guarantees, parent company guarantees) typically equal to ~£17,000–£25,000 per MW per year of remaining works, indexed; plus the new Progression Commitment Fee (PCF) under CMP448, payable on early termination/capacity reduction after Gate 2 acceptance Norton Rose Fulbright (Ofgem decision Q3 2025, implementation Q1 2026).
Typical end-to-end signed offer to energisation	Historically 4–7 years; under TMO4+ for Gate 2 projects, NESO is targeting “as soon as 2027” for Phase 1, but only for projects with a contracted connection date by end-2026 (clause 1 protected) or end-2027 (clause 2a protected). National Energy System Operator

Transmission-connected vs distribution-connected

- **Transmission-connected** (typically > ~50 MW, or any project requiring direct entry to the 132 kV+ network): NESO is the offer-issuing party; offers are termed BEGA/BELLA/CIA. Subject to TNUoS, Balancing Mechanism participation, BSC accession.
- **Distribution-connected (“embedded”)**: DNO issues the offer, with NESO providing the final Gate 2 determination above the relevant Transmission Impact Assessment (TIA) threshold (1 MW at most GSPs, though for some heavily-loaded GSPs NESO has agreed higher thresholds under CMP446). Embedded benefits (avoided TNUoS demand residual, GDUoS) have been largely eroded by reforms since 2018 but Triad-replacement and DUoS exposure remain.

Key documents and deliverables at this stage

Application form (CP47/CP14/CP15 family); Construction Agreement (BCA, MCUSA, BELLA, BEGA); System Impact Study / Connection Impact Assessment; design package to ER G99 standard; readiness evidence pack (lender’s expected diligence pack); securities/bank guarantees in place.

Common failure points

- Being assigned **Gate 1** rather than Gate 2 because planning is not yet submitted or land rights are insufficient — effectively a multi-year setback.
- **Strategic mis-alignment**: the Clean Power 2030 zonal capacities cap how much new solar can be accommodated regionally — projects in already-saturated regions (e.g. the East of England 132 kV network around Lincolnshire) face the highest risk.

- **Curtailment risk:** distribution networks increasingly require “active network management” (ANM) flexible/non-firm connections, which can curtail output 5–25% in early years.
- **NESO delays:** industry confidence in the December 2025 revised timetable is low; Ofgem in late 2025 was publicly “frustrated” with delays affecting 62% of Gate 2 project connections.

3. Planning and consenting

England — Town and Country Planning Act 1990 (≤ 100 MW from 31 December 2025)

From the implementation date of the Infrastructure Planning (Onshore Wind and Solar Generation) Order 2025 on **31 December 2025**, projects with a generating capacity **below 100 MW** in England are determined by the **Local Planning Authority (LPA)**. A one-year transitional window allows 50–100 MW projects already in pre-application to choose between the LPA route and the NSIP route until end-2026. For the decade prior, the threshold sat at 50 MW, producing the well-known “49.9 MW phenomenon”: Solar Energy UK confirmed in December 2024 that **174 projects had been built or were in the pipeline at 49.9 MW, and not a single one had been proposed between 50 MW and 99.9 MW** — evidence of the artificial cap.

Typical TCPA timeline and cost (2025/26):

Stage	Duration	Cost
Pre-application discussion with LPA, Statement of Community Involvement	3–6 months	£20,000–£80,000
EIA Screening & Scoping (Environmental Impact Assessment Regs 2017 — mandatory for solar >50 ha and routinely required above ~5–10 MW)	2–3 months	£15,000–£40,000
Environmental Statement preparation (full suite of surveys — see Section 4)	6–12 months	£150,000–£500,000

Application validation (LPA fee scales with site area)	2–4 weeks	LPA application fee typically £30,000–£60,000 for utility-scale; rises after 2023 fee review
Statutory determination ("8 weeks" / "16 weeks for EIA development") in practice	8–13 weeks if non-EIA; 6–18 months commonly with Planning Performance Agreements and extensions of time Solargridcheck for EIA projects	LPA Planning Performance Agreement (PPA) fees £20,000–£100,000
Section 106 negotiation	1–4 months	Legal fees £25,000–£75,000

Approval rate for solar farm applications at LPA level is **approximately 66%**

[Solargridcheck](#) (SolarGridCheck analysis of REPD; corroborated by ScienceDirect "Here comes the sun" 2024 study). Refusals can be appealed to the Planning Inspectorate (PINS) — appeal success rates for solar have historically been high (>50%) but add **9–18 months and £100,000–£300,000** in legal/consultant fees.

England — NSIP / Development Consent Order (≥ 100 MW from 31 December 2025)

For solar above 100 MW in England (or above 350 MW in Wales — but in practice almost no Welsh DCO solar exists), the project is consented as a **Nationally Significant Infrastructure Project** under the Planning Act 2008. [Norton Rose Fulbright](#) The Secretary of State for Energy Security and Net Zero (DESNZ) is the decision-maker; the **Planning Inspectorate (PINS)** examines.

The statutory DCO timetable is roughly:

Stage	Statutory minimum	Practical reality
Pre-application (scoping, statutory consultation under s42/47/48 etc.)	No statutory minimum but typically 18–30 months	Often longest stage; consultation under the Planning Act 2008 has now been streamlined by the 2025 Planning and Infrastructure Bill
Acceptance	28 days	28 days

Pre-examination	~3 months	~3 months
Examination	6 months	6 months
Recommendation (PINS to SoS)	3 months	3 months
Decision (SoS)	3 months	3 months (often slips; SoS recovery powers)
Legal challenge window	6 weeks	Now reduced under Banner Review reforms (Planning and Infrastructure Bill 2024/25) from three rounds to two

Total DCO timeline: typically 3.5–5 years from project inception to consent. The Government’s own operational review found that the time for an NSIP decision rose from **2.6 years in 2012 to 4.2 years in 2021.** [House of Commons Library](#) DCO costs for a large solar farm are routinely **£3–8 million**, dominated by legal, planning consultant, EIA and community consultation fees.

Recent DCO solar approvals include Cottam (600 MW, September 2024), West Burton (480 MW, January 2025 — and notably the only solar NSIP awarded a CfD in AR7a), Heckington Fen (500 MW, 2024), Sunnica (~500 MW, 2024), and Longfield (400 MW). Energy Secretary Ed Miliband undertook a “rapid-fire approval blitz” of solar NSIPs in summer 2024.

Wales — Developments of National Significance (DNS)

Projects **10–350 MW** in Wales follow the **Developments of National Significance (DNS)** route, decided by Welsh Ministers via Planning and Environment Decisions Wales (PEDW). DNS examinations are statutorily faster than DCOs (typically ~36 weeks from acceptance to recommendation). Solar above 350 MW in Wales would be a DCO; in practice no such project has yet been brought forward. Welsh Government policy under Future Wales: The National Plan 2040 generally supports large-scale renewables outside designated landscapes.

Scotland — Section 36 of the Electricity Act 1989

Onshore generating stations **over 50 MW** in Scotland (the threshold has **not** moved with the English reforms; a Scottish consultation in 2024/25 considered 100 or 150 MW)

[Brodies LLP](#) require **Section 36 consent from Scottish Ministers** via the **Energy Consents Unit (ECU)**, [Rtpi](#) with **deemed planning permission** granted under s.57(2) of the Town and Country Planning (Scotland) Act 1997. [Pas](#) Below 50 MW the local planning authority decides. [Dcslegal](#) Typical S36 timeline is **2–3 years** including the EIA report,

[Norton Rose Fulbright](#) and longer if a Public Local Inquiry is called (usually when the host council objects). Application fees are set by the Electricity (Applications for Consent and Variation of Consent) (Fees) (Scotland) Amendment Regulations 2022 [gov.scot](#) (variation applications currently incur no fee). [gov.scot](#) NPF4 (National Planning Framework 4) [Rtpi](#) heavily favours renewables, designating onshore wind and solar as “national developments.”

Northern Ireland

Under the Planning (Development Management) Regulations (NI) 2015: projects **under 5 MW** are LPA-determined; **5 MW and above** are “major developments” [Rtpi](#) decided by the Department for Infrastructure (12-week notice required). Article 39 of the Electricity (NI) Order 1992 provides a parallel consent regime, though the deemed planning provisions in the Electricity Consents (Planning) (NI) Order 2006 have not been brought into force.

[Norton Rose Fulbright](#) Marine licences (for any sub-sea elements) are issued by DAERA. Guidance suggests applications be submitted at least 4 months ahead of intended construction. [Norton Rose Fulbright](#)

Common reasons UK solar planning applications fail

In order of frequency (from REPD-based analysis and LPA officer reports):

1. **BMV land use** [Solargridcheck](#) without robust justification.
2. **Landscape and visual impact** — particularly proximity to National Landscapes (AONBs), AONB settings, and designated views.
3. **Heritage impact** — setting of listed buildings, conservation areas, scheduled monuments.
4. **Cumulative impact** with neighbouring solar/BESS schemes.
5. **Highway access** during construction (HGV movements).
6. **Glint and glare** affecting residences, aviation or transport.
7. **Inadequate BNG / ecological mitigation** under the post-2024 regime.
8. **Insufficient pre-application consultation** (especially fatal in DCO acceptance).

Key documents and deliverables at this stage

Planning Statement; Environmental Statement (ES); Design and Access Statement; Statement of Community Involvement; Heritage Impact Assessment; Landscape and Visual Impact Assessment; Flood Risk Assessment; Transport Assessment + Construction Traffic Management Plan; Biodiversity Net Gain Plan; Glint and Glare report; Outline Decommissioning and Restoration Plan; draft Section 106 agreement.

4. Environmental and technical studies

A standard utility-scale solar EIA / Environmental Statement covers the following workstreams. Costs below are for a 50 MW project; multiply by ~1.5–2.5x for a 100–500 MW DCO project.

Study	Purpose	Typical cost (50 MW)
Phase 1 Habitat / UK Habitat Classification (UKHab) survey	BNG baseline	£8,000–£20,000
Protected species surveys (bats, badgers, dormice, breeding birds, wintering birds, great crested newts via eDNA, reptiles, otters)	EPS licensing, planning condition discharge	£30,000–£100,000+ (seasonally constrained — bat surveys May–September only)
Biodiversity Net Gain (statutory 10% under Environment Act 2021, mandatory since Feb 2024) using the Natural England Biodiversity Metric	Demonstrating 10% net gain Lancs over 30 years, on-site or off-site units	Survey + metric £15,000–£40,000; off-site BNG units £25,000–£40,000 per “biodiversity unit” if purchased externally
Archaeology — desk-based assessment, geophysical survey, evaluation trenching	Heritage protection	£20,000–£150,000 (trenching is the big variable)
Heritage / Setting assessment	Impact on listed buildings, conservation areas, scheduled monuments	£10,000–£40,000
Flood Risk Assessment + Drainage Strategy (SuDS)	Sequential/exception test, NPPF para 173	£10,000–£30,000
Landscape and Visual Impact Assessment (LVIA) including ZTV, photomontages from agreed viewpoints	NPPF, GLVIA3 methodology	£25,000–£80,000
Glint and glare study	ICAO/CAA aviation safeguarding and residential receptors	£8,000–£20,000

Noise assessment (mainly for BESS and inverter compounds, ETSU-R-97/BS4142)	Residential amenity	£8,000–£25,000
Ground investigation (intrusive boreholes, trial pits, geotechnical, contamination Phase 2)	Foundation design, contaminated land	£30,000–£120,000
Topographic survey (LiDAR + ground truthing)	Layout design	£8,000–£25,000
Agricultural Land Classification (site-specific)	BMV justification	£5,000–£20,000
Traffic and Construction Traffic Management Plan	Highway authority	£10,000–£30,000
Community engagement (consultation events, leaflets, websites)	TCPA / DCO requirements	£20,000–£100,000+ (DCO scale)

Indicative all-in environmental study budget for a 50 MW TCPA project: £200,000–£500,000. For a 100–500 MW DCO project: £1.5–4 million.

In practice, solar projects routinely deliver BNG well above 10% — Botley West (the proposed 840 MW NSIP) claims 76%, [Coleshillsolarplans](#) RWE targets 50% as a corporate floor across UK sites, and British Solar Renewables targets 60%+ [Britishrenewables](#) — largely because converting intensively-cropped arable to managed grassland under panels is a relatively cheap, high-scoring intervention under the Natural England metric. Solar Energy UK’s “Solar Habitat 2023” pilot study of 87 operational solar farms found that managed sites support meaningful populations of red-listed species including skylark.

[Ecology by Design](#)

Common failure points at this stage

- Discovering Schedule 1 protected birds or roosting bat populations mid-EIA, forcing major redesign.
- Off-site BNG units unavailable in the LPA area, requiring expensive habitat banking.
- Late-stage Natural England objection on BMV grounds despite favourable LPA officer view.
- Aviation safeguarding objection (CAA / MOD / NATS) requiring panel re-orientation.

5. Design and engineering

Preliminary design and FEED

Design progresses in three rough stages:

1. **Concept / Pre-feasibility design** (Year 1): irradiance modelling (PVsyst, PVGIS) using TMY datasets, very coarse layout to support land option and grid application. **Cost: £15,000–£40,000.**
2. **Planning-stage design:** indicative layout, parameters (panel rows, inverter/transformer/BESS compound locations, cable corridors, access tracks, fencing, screening) — generally with a “Rochdale Envelope” approach giving design flexibility to substitute newer modules at construction. Single Line Diagrams to ER G99 level for the grid application.
3. **FEED / Detailed design / IFC** (Year 3–5, leading to EPC): bankable energy yield report (P50/P90/P99 by lender requirement, typically by DNV, Wood, Arup or RINA), final layout, geotechnical-driven foundation design (driven piles vs ballasted vs screw piles depending on chalk/clay/peat), HV single-line diagrams, protection coordination studies, SCADA architecture, BESS integration (if co-located). **FEED cost: £75,000–£300,000.**

Technology choices (2025/26 market reality)

- **Modules:** bifacial monocrystalline (typically 580–620 W per module from Tier-1 manufacturers — Jinko, Trina, LONGi, JA Solar — sometimes Canadian Solar or REC). Module cost has fallen sharply: forecast £200,000/MW for 2030 versus £270,000/MW in 2019. [Solar Power Portal](#)
- **Mounting:** in the UK the split has tipped toward **single-axis trackers** (which command ~£0.65–0.85/Wp EPC pricing) [IndexBox](#) for new utility-scale projects, expected to account for **55–60% of the EPC market.** [IndexBox](#) Fixed-tilt structures remain common for smaller (<20 MW) and constrained sites, at £0.55–0.75/Wp.
- **DC:AC ratio:** now routinely 1.3–1.5:1 (“overplanting”), supported by the 2024 High Court ruling confirming overplanting is not in principle inconsistent with national policy or sub-50 MW planning consent so long as effects are weighed sensibly.
- **Inverters:** central inverters (e.g. SMA, Sungrow, Power Electronics) for 100 MW+; string inverters increasingly favoured below ~50 MW.
- **HV plant:** typically 33 kV ring main units stepping to 132 kV or 400 kV at the developer’s substation; G99 compliance, grid forming inverters and reactive power

capability becoming standard requirements under the new Grid Code.

Key documents and deliverables at this stage

Bankable Energy Yield Report (P50/P90); Basis of Design; General Arrangement drawings; Single Line Diagrams (HV and LV); Substation layout; Cable Route Schedule; Foundation Design Report; BESS Integration Study (if applicable); SCADA & Communications Architecture; equipment data sheets (modules, inverters, trackers, transformers).

Common failure points at this stage

- Geotechnical surprises (running sand, peat, high water table, chalk solution features) requiring foundation redesign and cost overrun.
- Module/inverter selection becoming obsolete by the time of FEED, requiring substitution that triggers DNO/NESO compliance re-checks.
- Inability to fit overplanted capacity within Rochdale Envelope or planning permission's red line / panel array footprint.
- Lender Owner's Engineer rejecting P50/P90 yield assumptions used in financial model.

6. Offtake and route to market

Contracts for Difference — the dominant subsidised route

The Government's CfD scheme, run via the **Low Carbon Contracts Company (LCCC)**, provides a 15-year (now **20-year from AR7**) two-way price guarantee against an indexed strike price. Solar competes with onshore wind, hydro and other "Pot 1" mature technologies. Clearing strike prices for solar PV (normalised to 2024 prices where applicable; per LCP Delta and Energy UK analyses):

Round	Year	Solar capacity awarded	Solar strike price (2024 real)	Contract length
AR4	2022	2.2 GW (67 projects)	£64.09/MWh Solar Power Portal (originally £45.99/MWh in 2012 prices)	15 years
AR5	2023	1.9 GW (56 projects)	£65.49/MWh Solar Power Portal	15 years

AR6	September 2024	3.3 GW (93 projects) Solar Power Portal — record at the time	£72.92/MWh Solar Power Portal (≈ £61.53/MWh in 2012 real; ASP was £85/MWh in 2024 real)	15 years
AR7a	Results February 2026	4.9 GW (155 projects) Solar Power Portal — new record; average project size only 31.6 MW (the lowest in any round)	£65.23/MWh Solar Power Portal (10–12% below AR6); PV Tech ASP was £75/MWh, a 12% cut from AR6 reflecting module cost falls	20 years

Of the AR7a awards, 12 projects were in Wales and 11 in Scotland; West Burton (480 MW across three contracts) was the only NSIP-scale solar project to secure a CfD. AR7 marked a shift to 2024 real prices (previous rounds used 2012 prices) and to 20-year contracts, with LCP Delta consultants noting the lower strike price “indicates that the capital cost reductions of the past year are feeding through to the CfD.”

For comparison, Energy UK and DESNZ data show **AR6/AR7 solar strike prices of £65–73/MWh are well below the average wholesale price (>£80/MWh in 2025) and the cost of new-build gas (~£147/MWh).**

Corporate PPAs (CPPAs)

CPPAs remain the principal route for projects that do not secure a CfD. Key 2025/26 market characteristics (per Cornwall Insight, LevelTen, Pexapark and Renewable Exchange data):

- **Tenor:** typically **10–15 years**, with shorter (3–7 year) “tail” CPPAs increasingly used for repowered or post-RO assets, of which over 30 GW of RO-supported capacity is set to roll off support from 2027 onwards.
- **Pricing:** PPA prices have stabilised after months of European decline; UK utility-scale solar CPPAs are currently quoted in the **£55–£80/MWh range** as-generated. Pricing is now near the CfD clearing price, narrowing the merchant-vs-CfD trade-off. EDF Renewables UK have noted that corporate offtakers increasingly question paying above forward curve, with future power curves seen at ~£70/MWh.
- **Counterparties:** traditional utilities (EDF, Octopus, Statkraft, Centrica), increasingly **data centre and tech corporates** (Amazon, Microsoft, Google, Meta, the recently-announced 15-year TfL/EDF deal for an Essex solar project to be built in 2026), Bird & Bird and supermarkets/manufacturers (Tesco, Sainsbury’s, Lloyds Banking

Group's 10-year/50 GWh deal with Low Carbon). [Energy Institute](#)

- **Structures:** physical PPAs (sleeved supply), virtual / "synthetic" PPAs (CFDs financially settled), and pay-as-produced shapes are most common; pay-as-consumed structures rarer. CPPAs typically include **negative-price protection** (turn-down clauses), which CfDs do not [Drax](#) — a meaningful edge.
- **DESNZ "Call for Evidence on CPPAs"** (concluding July 2025) signalled potential further policy support, including consideration of a UK equivalent of France's Bpifrance €68m CPPA guarantee fund. [GOV.UK](#)

Merchant exposure and route-to-market

Most UK utility-scale solar projects financed today have a **blended revenue stack**: e.g. CfD on 70–80% of output for 15–20 years + merchant tail; or PPA on baseload-shape for 10 years + merchant on the rest; or floor-and-share structures with the offtaker. **Renewable Energy Guarantees of Origin (REGOs)** add £4–10/MWh; **Capacity Market** revenues are typically not available to solar in isolation but accessible via co-located BESS. Cornwall Insight and LCP Delta both publish quarterly forecasts of "captured price" — the volume-weighted price solar achieves, which is now ~85% of baseload due to cannibalisation, and forecast to fall further as installed solar capacity scales toward 50 GW.

Co-location with battery storage

Co-location is now the default design assumption for new build:

- **Penetration:** Industry analyses (Solar Media Market Research, Cornwall Insight, Aurora Energy Research) show **a large and growing share of new solar projects in planning are co-located with BESS**, often as "AC-coupled" arrangements sharing the grid connection and substation. Co-location attracts EPC pricing premia of ~£0.05–0.10/Wp for shared infrastructure.
- **Drivers:** capturing curtailment, capacity-market revenues, frequency-response (Dynamic Containment, Dynamic Moderation, Dynamic Regulation) and Balancing Mechanism revenues; smoothing solar revenue duck-curve cannibalisation; using "overplanted" inverter headroom.
- **Planning treatment:** BESS is determined by the LPA regardless of size [House of Commons Library](#) in England & Wales (it is not within the NSIP regime), but in Scotland BESS over 50 MW falls under S36. Co-located BESS within a DCO can be included in the DCO (e.g. Cleve Hill Solar Park).

Common failure points at this stage

- Failing to secure a CfD in a competitive auction (no second chance for 12+ months).
- CPPA counterparty credit downgrade (or insolvency) between LOI and signature — lenders insist on BBB- minimum, or parent company guarantee.
- Revenue stack assumptions in lender's base case undermined by Cornwall Insight / LCP Delta forecast updates showing accelerated cannibalisation.
- Falling foul of CfD eligibility conditions (e.g. supply chain plan, sustainable industry plan, Milestone Delivery Date).

7. Financing milestones

Development capital — at-risk spend through the stages

Development capital (the "at-risk" pre-FID spend) is borne entirely by the developer's equity. Indicative spend per MW, accumulated through stages (2025/26, 50 MW reference project):

Stage	Cumulative spend (£/MW)	Cumulative absolute spend (50 MW)
Origination (site identification, option signing)	£1,000–£3,000	£50,000–£150,000
Grid application accepted (pre-Gate 2 evidence)	£8,000–£15,000	£400,000–£750,000
EIA / planning application submitted	£15,000–£35,000	£750,000–£1.75m
Planning consent granted	£25,000–£55,000	£1.25m–£2.75m
Gate 2 grid offer accepted, securities lodged	£40,000–£70,000	£2.0m–£3.5m
CfD won OR PPA term sheet signed	£50,000–£80,000	£2.5m–£4.0m
EPC contract signed / Financial Close	£60,000–£100,000	£3.0m–£5.0m

For a **100 MW+ NSIP** the absolute spend is roughly proportionally higher in £m terms (because of DCO costs of £3–8m, larger BNG obligations, more extensive consultation) — typically **£8–15 million of at-risk capital to FID**, but per-MW £/MW is generally lower

(~£40,000–£70,000/MW) due to economies of scale on shared overhead.

Total project capex at financial close

DESNZ/Arup's July 2025 update to the levelised-cost assumptions puts current UK utility-scale solar PV all-in capex (total project cost including dev, EPC, owner's costs, financing) at the figures broadly consistent with WSP's 2024 work (£615/kW for an exemplar 20 MW plant in 2021 prices, [Solar Energy UK](#) escalated). The Renewable Energy Foundation/Hughes study found average capex of £0.95m/MW in 2018 prices [Ref](#) = **£1.16m/MW at 2023 prices** for plants built in 2017–2020, with limited further reductions since. EPC alone is now £550,000–£850,000/MW (see Section 8); total project capex is therefore in the **£800,000–£1.1m/MW** range for typical UK projects in 2025/26.

A 50 MW project therefore implies roughly **£40–55 million** total capex; a 100 MW project **£80–110 million**. The Arup 2024 study gave a medium LCOE for UK solar of **£46.5/MWh**, broadly consistent with the AR7a clearing strike of £65/MWh (which includes financing margin and merchant-tail discount).

Project finance structure

UK utility-scale solar (without a CfD) is typically financed as follows:

- **Capital structure:** 70:30 to 75:25 **non-recourse senior debt:equity** with a CfD; 60:40 to 65:35 for fully merchant or short-PPA projects.
- **Senior debt tenor:** 15–18 years amortising, often “soft mini-perm” with refinancing assumed at year 5–7.
- **Debt service coverage ratios:** 1.30–1.40x with a CfD, 1.40–1.60x for PPA-backed, 1.60x+ for merchant-heavy projects.
- **Lenders:** a deep market — NIB, NatWest, Lloyds, Santander, Triodos, ING, Rabobank, BNP Paribas, Société Générale, MUFG, SMBC, KfW IPEX; pension/insurance “long money” via private placements increasingly significant.
- **Equity:** dominated by infrastructure funds (Greencoat, Foresight, Bluefield, NextEnergy, Octopus, JLEN, Schroders Greencoat, Downing), pension funds (BBGI, USS), strategic developers (RES, BSR, Low Carbon, Enso, Innova, Island Green Power), and oil-major-backed platforms.
- **Tax:** equity returns benefit from R&D credits, capital allowances and (for some structures) Enterprise Investment Scheme/Venture Capital Trust availability. The 2024 ending of Agricultural Property Relief for solar-leased land has meaningfully changed landowner economics.

Key documents and deliverables at this stage

Information Memorandum; Lender's Information Pack; Financial Model (audited); Technical Due Diligence (Wood, DNV, Mott MacDonald); Legal Due Diligence Report; Term Sheet (debt and equity); Common Terms Agreement; Facility Agreements; Equity Subscription Agreement; Direct Agreements (DA with EPC, with offtaker, with O&M).

Common failure points at this stage

- Equity investor reset of return targets between MOU and financial close (especially in periods of rising base rates).
 - Lender requirement for additional debt service reserve, MRA reserve, decommissioning reserve eroding equity IRR.
 - Insurance market hardening (especially for hail, fire and BESS) forcing changes to project design or higher excess.
 - ECA or DFI cover (UKEF, BPI France) unavailable, narrowing the lender group.
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8. EPC contract

Tendering and selection

Most utility-scale UK solar projects run a competitive **two- or three-stage EPC tender**:

1. **EOI / RFI** to a long-list of 6–10 contractors (typically including ENERPARC, Atlantic Green, Bouygues E&S, Eelpower, Lightsource Construction Services, Conrad Energy Construction, RES Construction, Belectric, IB Vogt, GoldBeck Solar, RPS, McLaren Construction, Lark Energy).
2. **RFP** to a short-list of 3–4 with full BoQ, technical specifications and form of contract (commonly a heavily-amended FIDIC Silver Book for a full "wrap", or NEC4 ECC Option A/C for a split EPC).
3. **Negotiation and award**, with EPC signature typically conditional on financial close.

EPC pricing (UK, 2025/26)

Market evidence from IndexBox and industry sources for 2026:

Configuration	EPC price (£/Wp)	Implied £/MW (DC)
	£0.55–0.75/Wp	£550,000–

50 MW fixed-tilt, full wrap, no BESS	IndexBox	£750,000/MW
50 MW single-axis tracker, full wrap, no BESS	£0.65–0.85/Wp IndexBox	£650,000–£850,000/MW
100 MW+ tracker with co-located 2-hour BESS	£0.85–1.20/Wp (combined)	Variable

The UK ground-mount solar EPC market was estimated at **£1.8–2.2 billion in 2026** IndexBox (IndexBox), growing at ~7–9% CAGR to 2035; IndexBox **single-axis tracking represents 55–60%** of new build, with annual installed-capacity additions projected to rise from 3.5–4.5 GW per year in 2026 to 6–8 GW per year by 2035. IndexBox

Risk allocation and gatekeeping

A bankable EPC for UK solar will typically contain:

- **Lump-sum, date-certain, fully-wrapped scope** including all civils, mechanical install, HV/LV electrical, SCADA, commissioning, and grid energisation works.
- **Performance Liquidated Damages:** delay LDs ~0.1–0.2% of contract value per day, capped at 10–15%; performance LDs based on PR (Performance Ratio) and availability tests during a 2-year performance test period.
- **Total liability cap** typically 100% of contract value; LDs capped at 15–25%.
- **DNV/Wood/RINA Owner's Engineer** appointed by lenders.
- **Independent Engineer's certificates** required at milestone payments.
- **Defects liability period** 2 years, extended warranties on modules (12 years product, 25–30 years linear performance), inverters (10–15 years), trackers (5–10 years).
- **Split EPC** (separate "supply" + "install" contracts) is common to optimise VAT and currency risk on imported modules; it requires a robust interface agreement and a wrap from an investment-grade contractor or coverage by a separate interface insurance product.

What must be in place before EPC signature

The classic four-corner gate to FID — these are non-negotiable conditions precedent to drawdown for almost any UK lender:

1. **Land secured** — option exercised into a registered long-term lease (or freehold), with all easements (cable, access, substation), wayleaves for the grid connection cable route,

and rights of way regularised. Vacant possession achievable on the relevant commencement date.

2. **Planning consent (DCO, TCPA permission or S36 + deemed PP)** — non-judicially-reviewable (i.e. **6-week JR window expired** with no challenge, or any challenge resolved). All pre-commencement conditions discharged or capable of discharge prior to construction start.
3. **Accepted grid connection offer** with confirmed Gate 2 status under TMO4+, securities lodged, and a connection date consistent with the EPC programme (typically within 18 months of mechanical completion).
4. **Offtake / route to market** — either an executed CfD with LCCC and a Milestone Delivery Date deliverable, or an executed long-term CPPA from a bankable counterparty (BBB- or better, or appropriate credit support), or a hedged merchant strategy acceptable to lenders.
5. **Insurance package** in place (CAR/EAR, business interruption, public liability, professional indemnity from designers).
6. **Equity funded or committed** via equity bridge / cash equity, and senior debt facility agreements signed.
7. **EPC contractor selected**, with parent company guarantee, performance bond (typically 10% of contract value) and retention.

Key documents and deliverables at this stage

EPC Contract; Parent Company Guarantee; Performance Bond; Retention Bond; Direct Agreement with lenders; O&M Contract (separate or sub-contracted from EPC); Module Supply Agreement (if split EPC); Inverter Supply Agreement; Long-Lead Item POs (transformers, switchgear); Independent Engineer's Bankability Report; Insurance certificates.

Common failure points at EPC stage

- **Module supply chain risk** — UK Solar Stewardship Initiative compliance (Uyghur Forced Labour Prevention Act / UK Modern Slavery Act), pushing developers to traceable supply.
- **Late grid energisation date** clashing with EPC programme.
- **Cost inflation** between FEED and EPC tender (steel, copper, transformer lead times now 18–24 months for large units).
- **Lender conditions precedent** not satisfied — most often planning condition discharge

or land registry issues on cable easements.

- **EPC contractor balance sheet** — many specialist solar EPCs have constrained balance sheets, prompting lenders to demand strong parent guarantees and on-demand performance bonds.

9. REPD-derived UK solar project data

The **Renewable Energy Planning Database (REPD)**, managed by Barbour ABI on behalf of DESNZ and updated quarterly [GOV.UK](#) (most recent published edition January 2025), is the canonical record of UK renewables projects above 150 kW. [GOV.UK](#) The user-supplied REPD CSV (~4.6 MB, hosted at github.com/VentusLtd/globalgrid2050/blob/main/repd.csv) was too large to render directly in this session, but its content is well-characterised by published analyses based on the same data, and the data.barbour-abi.com/smart-map/repd/desnz/ interactive viewer and gov.uk publication summarise it quarterly:

- **Typical project size:** until 2024 the modal utility-scale UK solar project was **49.9 MW** — Solar Energy UK confirmed **174 projects either built or in the pipeline at 49.9 MW versus zero in the 50–99.9 MW band**, an artefact of the (old) NSIP threshold. Above this, the pipeline jumps to NSIP-scale: a cluster around 350–600 MW (Sunnica, Cottam, Longfield, Heckington Fen, West Burton, Botley West, Mallard Pass). The AR7a results show average winning solar project size of just **31.6 MW** — the lowest of any round.
- **Approval rates at LPA level:** ~66% of solar farm applications are approved [Solargridcheck](#) on first determination [Solargridcheck](#) (SolarGridCheck/REPD; corroborated by ScienceDirect “Here comes the sun” 2024 study using REPD plus planning officer reports, which finds that approval likelihood is strongly driven by planning officer recommendation and the absence of statutory consultee objection rather than by raw public-comment volume).
- **Time from application to consent (TCPA route, EIA development):** typically **12–18 months**, with significant variance — straightforward sub-EIA applications can be 6–8 months, while heavily contested sites can run 24–36 months with extensions, callbacks and appeals.
- **Time from consent to construction start:** typically **18–36 months**, reflecting grid energisation date dependency, EPC tendering, and equity/debt close. The Heckington Fen experience (delayed by land-rights disputes with the Crown Estate and Duchy of Lancaster) shows how easily this can drift to 4–5 years.
- **Attrition between development stages** (typical industry benchmark, broadly consistent with REPD status-change data):

- Sites identified → option signed: ~40–50% (developer pipelines routinely look at 4–5x more sites than they option).
- Option signed → grid offer accepted: ~50–70%.
- Grid offer → planning submitted: ~70–85%.
- Planning submitted → planning consent (first determination): ~66%.
- Consent → financial close: ~75–85%.
- **Overall: roughly 8–15% of greenfield sites identified ever reach EPC signature.**
This attrition rate is a key reason why developer-stage equity returns are higher than operational solar equity returns.

10. Indicative end-to-end timelines

A 49.9 MW TCPA route project

Quarter	Activity
Q1 Year 1	Site identification, Heads of Terms, option signing
Q3 Year 1	Grid application submitted (now under TMO4+ gated window)
Q1 Year 2	EIA scoping, environmental surveys commence
Q3 Year 2	Pre-application consultation; Gate 1 → Gate 2 readiness
Q1 Year 3	Planning application submitted
Q4 Year 3	Planning consent (assuming approval at first determination)
Q2 Year 4	CfD bid (AR8 / AR9) or CPPA term sheet; Gate 2 offer accepted
Q4 Year 4	Financial close + EPC contract signature
Q4 Year 5 / Q1 Year 6	Commercial Operation Date (COD)

Total: ~4 years from site identification to EPC signature; 5–6 years to COD.

A 200 MW NSIP / DCO project

Quarter	Activity
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Q1 Year 1	Site identification, multi-landowner option strategy
Q1 Year 2	Transmission grid application; EIA Scoping Opinion
Q3 Year 2	Statutory pre-application consultation (s42/47/48) — round 1
Q3 Year 3	Statutory consultation round 2 + Preliminary Environmental Information
Q3 Year 4	DCO application submitted
Q4 Year 4	DCO accepted; pre-examination
Q2–Q4 Year 5	Examination (6 months)
Q1 Year 6	PINS recommendation; SoS decision (3+3 months)
Q3 Year 6	DCO granted (subject to 6-week JR window)
Q1 Year 7	Financial close + EPC contract signature
Q4 Year 8	COD

Total: ~6–7 years from site identification to EPC signature; 8 years to COD. The fast-track DCO route (PINS, since 2024) can compress this by 4–6 months for qualifying projects.

11. Cumulative capital at risk by stage — summary table

For a typical **50 MW TCPA-route project** in 2025/26 (figures are indicative; multiply by ~1.5–2x for 100–250 MW DCO projects, less per-MW but more in absolute terms):

Cumulative milestone	Cumulative dev capital at risk (£)	£/MW	Key gating documents
Option signed, HoTs concluded	£50k–£150k	£1k–£3k	Option Agreement, Heads of Terms
Grid connection application accepted (under TMO4+ window)	£400k–£750k	£8k–£15k	DNO/NESO offer, connection studies
EIA Scoping Opinion +	£750k–	£15k–	EIA Scoping report, full

planning application submitted	£1.75m	£35k	Environmental Statement, Design & Access Statement
Planning consent granted (consent + JR window expired)	£1.25m–£2.75m	£25k–£55k	Planning permission, Section 106 agreement, BNG Plan, condition compliance plan
Gate 2 offer accepted, securities posted	£2.0m–£3.5m	£40k–£70k	Bilateral Connection Agreement (BCA) or DNO Connection Offer, security/PCB
Offtake secured (CfD award OR PPA term sheet)	£2.5m–£4.0m	£50k–£80k	CfD contract w/LCCC OR signed PPA term sheet
EPC contract signed / Financial Close	£3.0m–£5.0m	£60k–£100k	EPC contract, equity & debt agreements, insurance, OE reports, lender CPs satisfied
Construction (~12 months) — additional capex	+ £30–45m total project capex	£600k–£900k/MW	—
COD	Project operational	—	Acceptance certificates, taking-over certificate

For a **200 MW DCO project**, multiply the dev-capital row at EPC signature by ~3–4x in absolute terms (£10–15m total at-risk dev capital, including £3–8m on the DCO process itself), though the **per-MW figure typically falls to £40,000–£70,000/MW** owing to scale.

12. Closing observations

UK solar development in 2025/26 sits at an inflection point. The **NSIP threshold increase to 100 MW**, the **TMO4+ “first ready, first connected” connection regime**, the new **20-year CfD contracts**, the **record 4.9 GW awarded in AR7a at £65/MWh** and the **Clean Power 2030 target of 45–57 GW** together offer a structurally more favourable environment than at any point since the Renewables Obligation peak of 2014–16. Nonetheless, the binding constraints have shifted: where once it was subsidy availability, today the principal gatekeepers are **grid (specifically Gate 2 status and connection date)**, **planning (specifically BMV land and LVIA)**, and **offtake-vs-merchant economics**. Cost discipline — module prices below £200,000/MW, EPC pricing of £550,000–£850,000/MW, total project capex below £1.1m/MW — is necessary but no longer

sufficient. The successful developer in 2026 is one who can sequence land, grid, planning and offtake so that each gate clicks shut on time, allowing the project to reach EPC signature with around £60,000–£100,000/MW of equity at risk, before the larger £600,000+/MW EPC commitment is made. Roughly only one in eight greenfield options gets that far.

Forecasts and forward-looking figures in this report (especially future strike prices, CPPA price ranges, market-size projections, Aurora/Cornwall Insight captured-price assumptions, and IndexBox's installed-capacity additions out to 2035) reflect industry consensus at time of writing and should be treated as scenario rather than as fixed values. Connection timetables under TMO4+, in particular, remain subject to NESO's evolving programme, with several already-announced delays since the original June 2025 go-live.